

Explainable Deep Learning for Multi-Modal Medical Imaging

Toward Trustworthy AI-Assisted Diagnosis

MedDiagnosis

MedDiagnosis - Research Prototype & Implementation Report

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This document describes the motivation, methodology, architecture, and implementation of an explainable AI framework for medical image analysis. It is intended for academic review and project guide presentation. Research and educational use only - not for clinical diagnosis.

Abstract

Deep learning has achieved strong performance in medical image analysis, yet black-box predictions limit clinical adoption because clinicians require transparent, verifiable reasoning. This project addresses the need for trustworthy AI-assisted diagnosis by developing MedDiagnosis, an explainable deep learning framework for multi-modal medical imaging. The system integrates a biomedical Vision-Language Model (VLM) with visual attribution (Grad-CAM / input-gradient saliency), natural-language report generation, and a faithfulness evaluation module that checks whether textual explanations align with model focus regions. The current prototype targets radiology (chest X-ray analysis) as the primary modality, with a modular architecture designed to extend toward digital pathology and broader multi-modal fusion. A Gradio interface and CLI enable interactive demonstration, batch evaluation on a curated test dataset, and follow-up clinical-style question answering.

1 Introduction and Background

Medical imaging underpins modern diagnosis in radiology (X-ray, CT, MRI) and digital pathology (whole-slide histopathology). Convolutional neural networks and, more recently, vision-language models can detect abnormalities, generate reports, and answer visual questions. However, regulatory bodies and clinical workflows demand interpretability: clinicians must understand why a model reached a conclusion before trusting it.

Explainable AI (XAI) methods such as Grad-CAM, saliency maps, and attention visualization localize model decisions in image space. In radiology, this supports lesion localization and report verification. In digital pathology, similar techniques can highlight malignant regions on H&E slides. Multi-modal systems that combine imaging with text prompts or clinical context further improve usability but increase the need for cross-modal faithfulness - ensuring language explanations reflect visual evidence.

MedDiagnosis implements this vision as a research prototype: a lightweight, locally runnable pipeline that demonstrates end-to-end AI-assisted chest X-ray analysis with built-in explainability and quantitative grounding checks.

2 Problem Statement

Despite advances in medical deep learning, three barriers prevent trustworthy deployment:

- Opacity: Deep models provide predictions without showing which image regions drove the decision.
- Unverified language: VLMs can produce plausible reports not grounded in visual evidence.
- Modality silos: Radiology and pathology tools are often separate, limiting unified XAI frameworks.

The core problem this project tackles is: How can we build an explainable, multi-modal medical imaging system that generates diagnostically useful outputs while providing visual and textual evidence clinicians can inspect and evaluate?

3 Objectives and Scope

3.1 Primary Objectives

- Design an explainable deep learning pipeline for medical image diagnosis.
- Integrate a biomedical VLM for radiology report generation and visual question answering.
- Apply visual attribution (Grad-CAM / saliency) to localize model focus on the input image.
- Evaluate explanation faithfulness by comparing anatomy terms in text with heatmap regions.
- Provide an interactive demo (Gradio UI) and batch evaluation on a test dataset.

3.2 Project Scope

- Implemented: Chest X-ray (radiology) analysis using BMC-SmolVLM-256M on local Mac/GPU.
- Implemented: XAI layer, faithfulness scoring, CLI, Gradio UI, 8-image test dataset.

- Planned extension: Digital pathology (H&E whole-slide patches) using the same VLM + XAI pipeline.
- Planned extension: Multi-modal fusion (image + clinical notes / lab values) in future phases.

Scope note: The current codebase is a functional radiology prototype. Digital pathology is addressed architecturally and in future work; it is not yet implemented in code.

4 Literature Context

This work sits at the intersection of medical AI, explainability, and vision-language modeling:

- Medical VLMs (e.g., MedGemma, PMC-VQA, BIOMEDICA models) combine image understanding with natural-language generation for radiology and biomedical tasks.
- Grad-CAM (Selvaraju et al.) and gradient-based saliency provide class-discriminative localization in CNNs and vision encoders.
- Faithfulness metrics in XAI assess whether explanations reflect model behavior (same decision under perturbation) and whether text matches saliency.
- Benchmarks such as VQA-RAD, MIMIC-CXR, and ChestX-ray14 support evaluation of visual question answering and report generation in radiology.
- Digital pathology XAI (patch-level classification + heatmaps on WSIs) parallels radiology attribution but operates at cellular/tissue scale.

MedDiagnosis contributes a practical, lightweight implementation that unifies VLM inference, visual attribution, and a simple faithfulness heuristic suitable for academic demonstration on consumer hardware (Apple MPS / CPU).

5 Proposed Framework

The proposed trustworthy AI-assisted diagnosis framework consists of four layers:

```

+-----+
| Layer 4: Trust & Evaluation |
| Faithfulness score, anatomy grounding, batch metrics |
+-----+
| Layer 3: Explainability (XAI) |
| Grad-CAM, input-gradient saliency, heatmap overlays |
+-----+
| Layer 2: Multi-Modal Reasoning (VLM) |
| Image + prompt -> findings, diagnosis, VQA responses |
+-----+
| Layer 1: Input Modalities |
| Radiology (CXR) [implemented] | Pathology [planned] |
+-----+

```

5.1 Radiology Modality (Current Implementation)

Chest X-rays are ingested as RGB images. A structured or free-text prompt (e.g., full report, findings only, VQA question) is passed to the VLM. The model generates findings and diagnosis text. Optionally, the vision encoder is probed for attribution maps overlaid on the original radiograph.

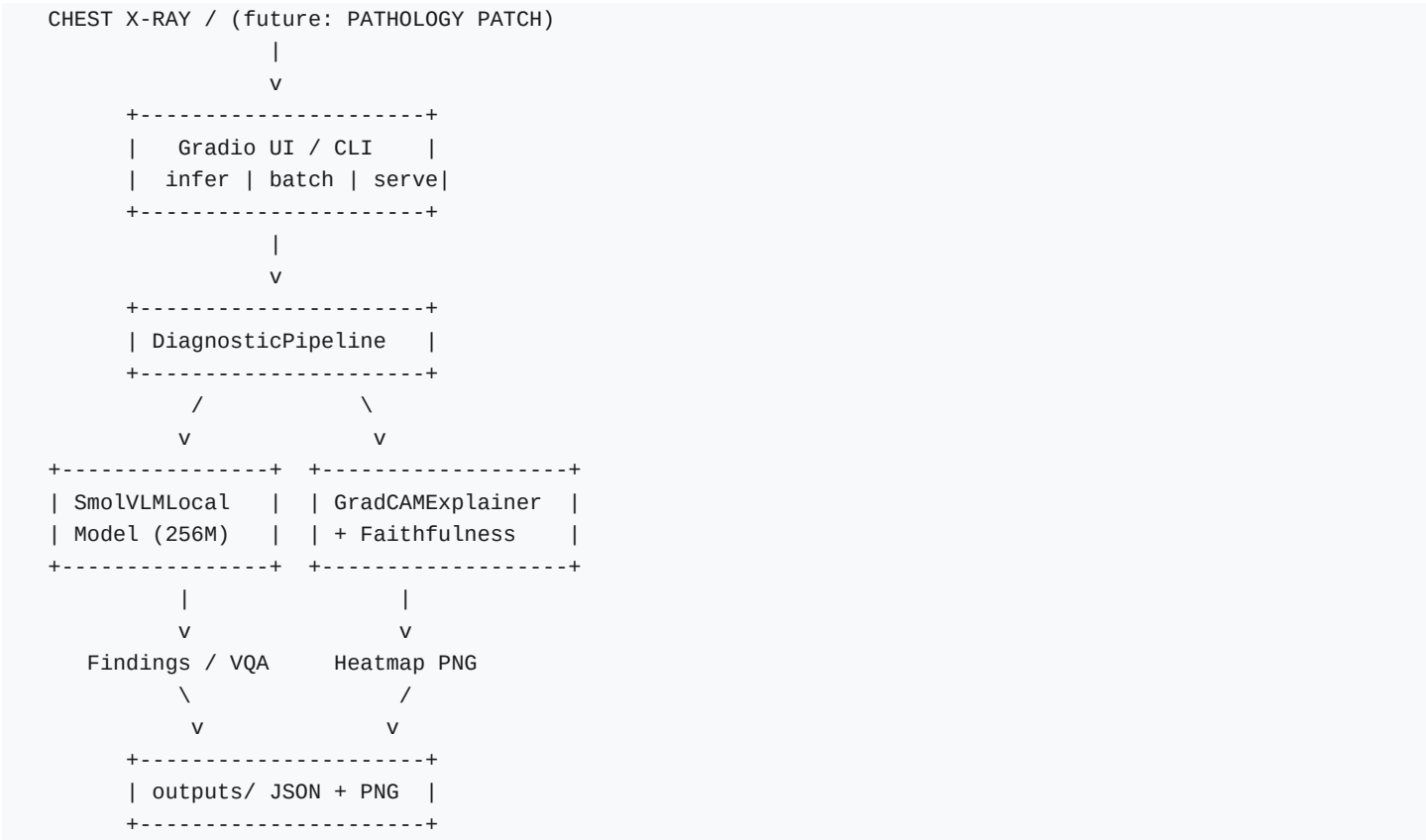
5.2 Digital Pathology Modality (Planned Extension)

The same pipeline architecture applies to histopathology: patch or tile images from H&E-stained slides are fed to the VLM with prompts such as 'Describe tissue morphology' or 'Is malignancy present?'. Grad-CAM highlights regions of abnormal cellularity. Modularity in DiagnosticPipeline allows swapping modality-specific prompts and anatomy/tissue region maps for faithfulness evaluation.

5.3 Trustworthiness Criteria

- Transparency: User sees both text output and visual heatmap.
- Grounding: Faithfulness module checks text-image alignment.
- Human-in-the-loop: Gradio chat allows clinician-style follow-up questions.
- Reproducibility: Config-driven pipeline, saved JSON/PNG outputs per sample.

6 System Architecture



6.1 Data Flow

- User uploads image via CLI (infer/batch) or Gradio web UI.
- Image and prompt are formatted via SmolVLM chat template and tokenized.
- BMC-smolvlm1-256M generates radiology findings (default max 128 tokens).
- If XAI enabled: input-gradient saliency (Mac MPS/CPU) or Grad-CAM (CUDA) produces heatmap.
- Faithfulness module extracts anatomy terms from text and maps heatmap center to lung/heart regions.
- Results saved to outputs/ as JSON and attribution PNG; Gradio supports follow-up chat.

6.2 Technology Stack

Language:	Python 3.10+
Deep Learning:	PyTorch, Hugging Face Transformers
Model:	BIOMEDICA/BMC-smolvlm1-256M (256M params, biomedical VLM)
Processor:	HuggingFaceTB/SmolVLM-256M-Instruct
XAI:	Grad-CAM, input-gradient saliency, OpenCV heatmap overlay
UI:	Gradio 5+ (optional [ui] install extra)
Runtime:	Apple MPS / CUDA / CPU (~500 MB model, runs locally)

7 Implementation Details

7.1 Project Structure

meddiagnosis/	
config/default.yaml	Model, prompts, XAI settings
data/test_cxr/	Test dataset (8 chest X-rays + manifest)
src/meddiagnosis/	
cli.py	infer, batch, serve commands
config.py	YAML config loader
data/test_dataset.py	TestDataset loader from manifest
models/smolvlm.py	Local VLM wrapper
models/vlm_common.py	Chat template, tokenization, generation
pipeline/inference.py	DiagnosticPipeline orchestrator
xai/gradcam.py	Grad-CAM + fast saliency (Mac)
xai/visualization.py	Heatmap overlay rendering
evaluation/faithfulness.py	Anatomy grounding score
app/gradio_app.py	Web UI + follow-up chat
scripts/download_test_dataset.py	
scripts/generate_project_pdf.py	
outputs/	JSON results + XAI PNGs

7.2 Key Modules

- SmolVLMLocalModel: Wraps BMC-smolvlm1-256M; supports generate(), chat(), prepare_inputs() for XAI.
- DiagnosticPipeline: Orchestrates inference, optional XAI, faithfulness, and JSON export.
- GradCAMExplainer: Hook-based Grad-CAM on CUDA; fast input-gradient saliency on MPS/CPU.
- FaithfulnessEvaluator: Maps heatmap focus to anatomy regions; compares with terms in generated text.
- TestDataset: Loads manifest.json for batch evaluation with category labels and VQA questions.
- Gradio App: Prompt presets, Grad-CAM toggle, example images, interactive follow-up Q&A.

8 Explainability Methods

8.1 Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM computes a coarse localization map by weighting forward-pass activations of the vision encoder with gradients of the target output (generated token logits) with respect to those activations. The resulting heatmap highlights regions that most influenced the model's response. On CUDA GPUs, full Grad-CAM with backward hooks is used.

8.2 Fast Input-Gradient Saliency (Mac / CPU)

Apple MPS backward passes are slow (~2-3 minutes). When config `xai.fast` is true, the system uses input-gradient saliency: gradients of the output with respect to input pixels approximate important regions. This trades some spatial precision for practical runtime on Mac hardware while preserving explainability for demonstration.

8.3 Visualization

Heatmaps are normalized, color-mapped (JET), and alpha-blended onto the original X-ray using OpenCV. Overlays are saved to `outputs/xai/<sample-id>_gradcam.png` for inspection and inclusion in project reports.

9 Faithfulness Evaluation

Trustworthy AI requires that natural-language explanations reference the same anatomical regions the model visually attended to. The faithfulness module implements a lightweight grounding check:

- Extract anatomy terms from generated text (e.g., right upper lung, heart, pleura).
- Compute center-of-mass of the saliency heatmap in normalized image coordinates.
- Map heatmap focus to predefined anatomical bounding boxes for chest X-rays.
- Faithfulness score = 1.0 if predicted focus region appears in mentioned terms, else 0.0.

This heuristic provides a quantitative sanity check for demo and research purposes. Future work may integrate Chest ImaGenome annotations or human expert ratings for more rigorous evaluation.

10 Dataset and Experimental Setup

10.1 Test Dataset

A curated test set of 8 chest X-ray images is stored in `data/test_cxr/images/` with metadata in `manifest.json`. Sources include Wikimedia Commons (CC0 / public domain) and VQA-RAD chest samples (research use).

- 3 normal chest X-rays (PA and standard views)
- 2 pneumonia cases (lobar pneumonia, AP view with upper lobe opacity)
- 3 VQA-RAD chest images with reference questions and answers

10.2 Download and Batch Evaluation

```
python scripts/download_test_dataset.py
python -m meddiagnosis.cli batch --no-xai --max-tokens 80
python -m meddiagnosis.cli batch --limit 3 # subset
```

10.3 Evaluation Metrics

- Qualitative: Visual inspection of findings text and Grad-CAM overlays.
- Faithfulness score: Binary grounding alignment per sample.
- Latency: Model load time (~10s), inference (~4-5s), XAI (~2-3 min on Mac MPS).
- Future: VQA accuracy against VQA-RAD answers, BLEU/ROUGE for report generation.

11 Demo Workflow and Usage

11.1 Installation

```
python -m venv .venv && source .venv/bin/activate
pip install -e ".[ui]"
```

11.2 Single-Image Inference

```
# Fast (no XAI)
python -m meddiagnosis.cli infer \
  --image data/test_cxr/images/cxr_002_lobar_pneumonia.jpg \
  --no-xai --max-tokens 80

# With explainability
python -m meddiagnosis.cli infer \
  --image data/test_cxr/images/cxr_002_lobar_pneumonia.jpg \
  --max-tokens 80
```

11.3 Interactive Web Demo

```
python -m meddiagnosis.cli serve --port 7860
# Open http://127.0.0.1:7860
# Upload X-ray, select prompt preset, toggle Grad-CAM, ask follow-up questions
```

11.4 Outputs

- outputs/<sample-id>_result.json - findings, prompt, gradcam_path, faithfulness dict
- outputs/xai/<sample-id>_gradcam.png - heatmap overlay on input image

12 Research Contribution

This project demonstrates a practical path toward trustworthy AI-assisted diagnosis by combining:

- Pre-trained biomedical VLM (no full model training required; suitable for M.Tech timeline).
- Multi-level explainability: visual attribution + language explanation + grounding metric.
- Modular architecture extensible from radiology to digital pathology and multi-modal inputs.
- Local, reproducible deployment on consumer hardware for academic demonstration.

The contribution is methodological and engineering: a unified XAI framework prototype that a project guide can evaluate through live demo, batch runs, and saved attribution artifacts.

13 Future Work

- Digital pathology: Apply pipeline to H&E patch datasets (Camelyon, PatchCamelyon).
- Multi-modal fusion: Incorporate clinical notes, lab values, or structured EHR fields.
- Stronger evaluation: VQA-RAD accuracy, expert radiologist review, Chest ImaGenome grounding.
- Model upgrades: MedGemma 4B or domain-fine-tuned CXR models when GPU resources allow.
- Integrated Grad-CAM variants: LayerCAM, Score-CAM for comparison studies.
- Regulatory awareness: Document bias, uncertainty quantification, and audit trails.

14 Limitations and Ethics

- Research prototype only - not validated for clinical use or regulatory approval.
- BMC-smolvlm trained on general biomedical literature, not chest X-rays specifically.
- Faithfulness uses heuristic anatomy boxes, not expert-annotated ground truth.

- Digital pathology modality is planned but not yet implemented.
- XAI backward pass remains slow on Apple MPS; full Grad-CAM preferred on CUDA.
- Dataset is small (8 images); conclusions are illustrative, not statistically robust.
- AI outputs must not replace qualified medical professionals.

15 Conclusion

MedDiagnosis implements an explainable deep learning framework aligned with the theme 'Explainable Deep Learning for Multi-Modal Medical Imaging: Toward Trustworthy AI-Assisted Diagnosis in Radiology and Digital Pathology.' The radiology prototype successfully combines VLM-based report generation, visual attribution, and faithfulness evaluation in an interactive, reproducible system. The modular design provides a clear foundation for extending to digital pathology and richer multi-modal inputs in subsequent project phases. This document and the accompanying codebase support demonstration and review by the project guide.

References (Selected)

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